Systems Analysis and Control

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Lecture 15: Root Locus: Computing Gains and Generalized Root Locus

Overview

In this Lecture, you will learn:

Recovering the Gain, k, from root locus plot.

- Picking Points
- Calculating the Gain
- Satisfying Performance Criteria

Generalized Root Locus?

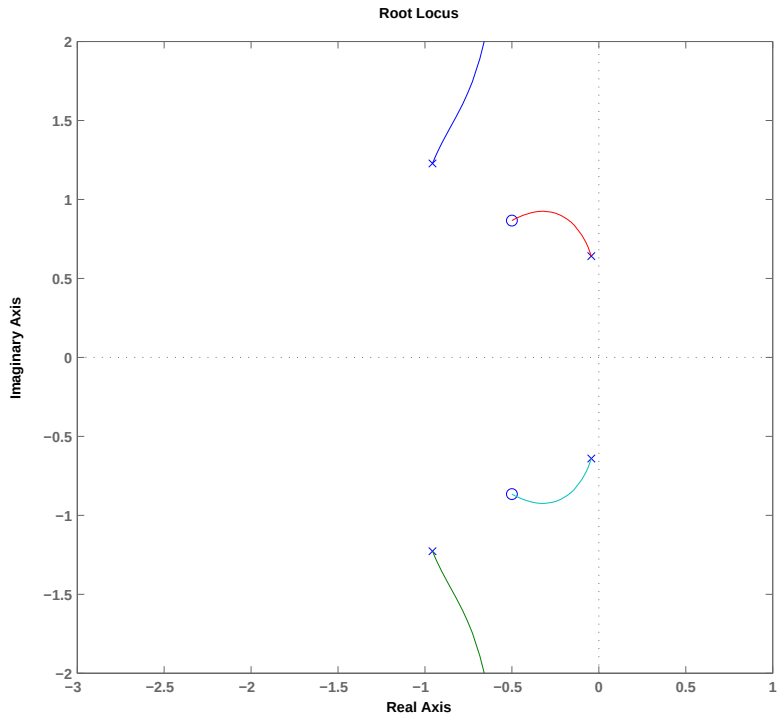
- What about changing *OTHER* parameters
- T_D, T_I, et c.
- mass, damping, et c.

Calculating Gain at a Point on the Root Locus

Suppose we have drawn our root locus.

Now we want:

- A point with 20% overshoot
- A point with 4s settling time
- A point with 2s rise time.



We can see that acceptable points are on the root locus.

Question: What value of k will achieve these points?

Calculating Gain at a Point on the Root Locus

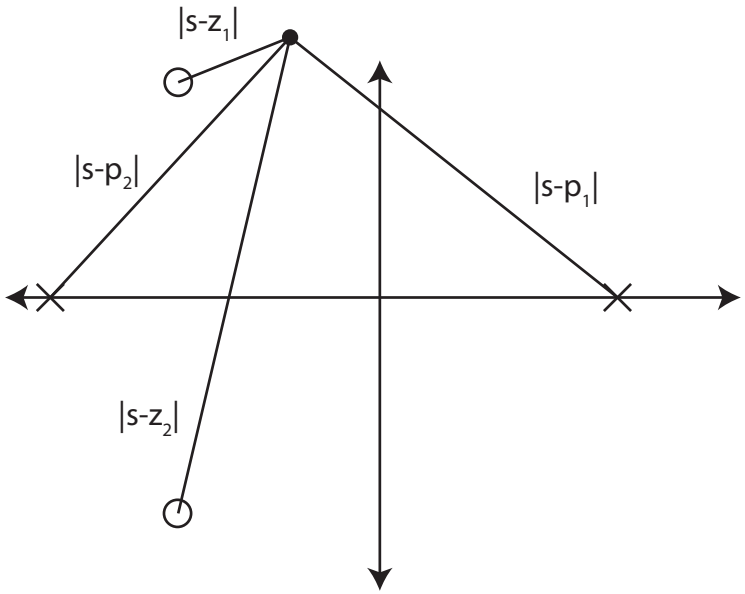
Problem: Given a point on the root locus, $s_{desired}$, find the gain which achieves that point.

Answer: We know that for a point on the root locus,

$$1 + kG(s) = 0$$

Therefore, the gain at the point $s_{desired}$ is

$$k = \left| -\frac{1}{G(s)} \right| = \frac{1}{|G(s)|}$$



- The Gain, k, at point, s, is the *magnitude* of $G(s)$.

Note: For a point *NOT EXACTLY* on the root locus, this still gives an approximate gain

Formula Gain at a Point on the Root Locus

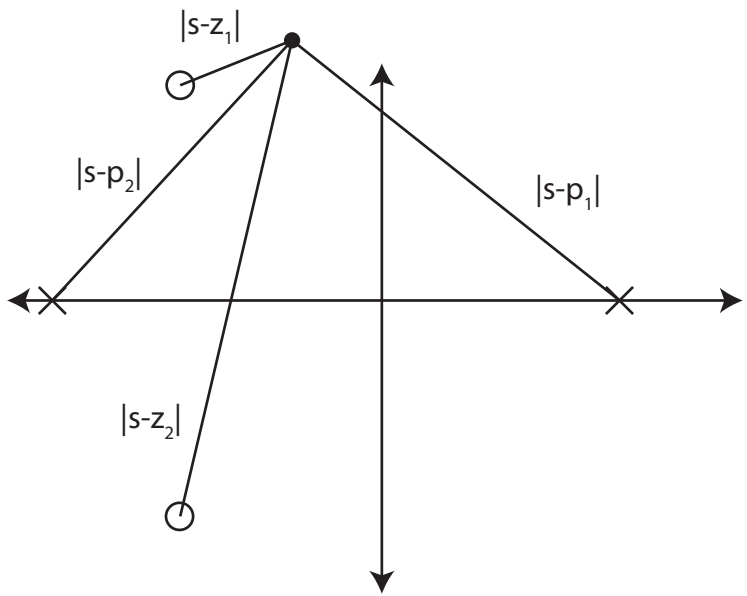
Exploiting multiplication/division rules for polar form of complex numbers

Calculating the magnitude of $G(s)$ is similar to calculating the phase

$$G(s) = \frac{(s-z_1)\cdots(s-z_m)}{(s-p_1)\cdots(s-p_n)} \qquad k = \frac{1}{|G(s)|}$$

Multiplication and division properties of complex numbers:

$$|r_1 \cdot r_2| = |r_1| \cdot |r_2| \qquad \left|\frac{r_1}{r_2}\right| = \frac{|r_1|}{|r_2|}$$



To calculate $|G(s)|$ we can use

$$|G(s)| = \frac{|s-z_1|\cdots|s-z_m|}{|s-p_1|\cdots|s-p_n|}$$

$$k(s) = \frac{|s-p_1|\cdots|s-p_n|}{|s-z_1|\cdots|s-z_m|}$$

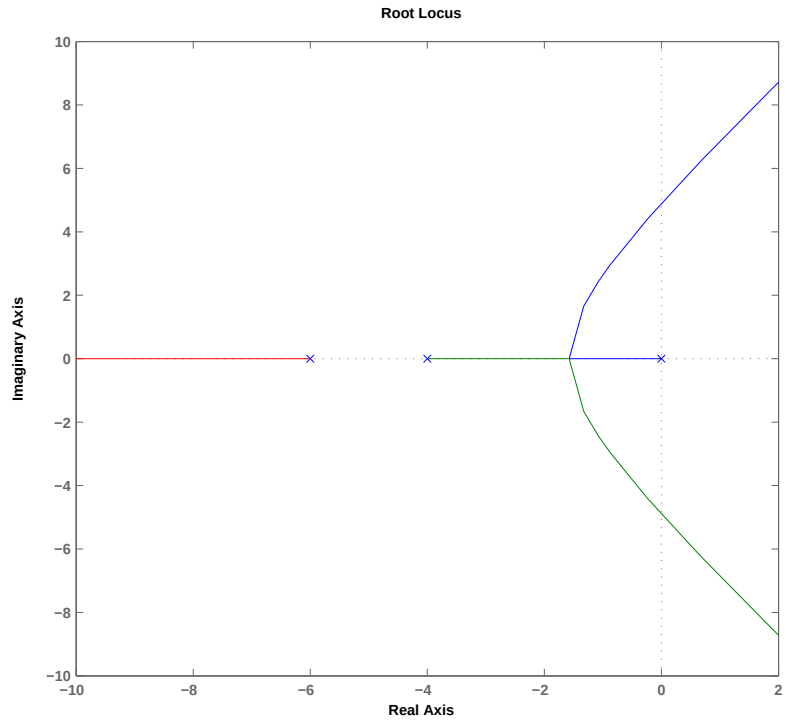
Calculating Gain at a Point on the Root Locus

Numerical Example: a point on the real axis

Suppose we have factored the transfer function as.

$$G(s) = \frac{1}{s(s+4)(s+6)}$$

Find the gain at break point $s = -1.8$.



Magnitude from Pole 1:

$$|s| = 1.8$$

Magnitude from Pole 2:

$$|s+4| = 2.2$$

Magnitude from Pole 3:

$$|s+6| = 4.2$$

$$k = \frac{|s-p_1| \cdots |s-p_n|}{|s-z_1| \cdots |s-z_m|} = \frac{1.8 \cdot 2.2 \cdot 4.2}{1} = 16.63$$

Calculating Gain at a Point on the Root Locus

Numerical Example: A point NOT on the real axis

Points on the real axis are easiest. Lets try the point at $s \cong -1 + 2\imath$

$$G(s) = \frac{1}{s(s+4)(s+6)}$$

Magnitude from Pole 1:

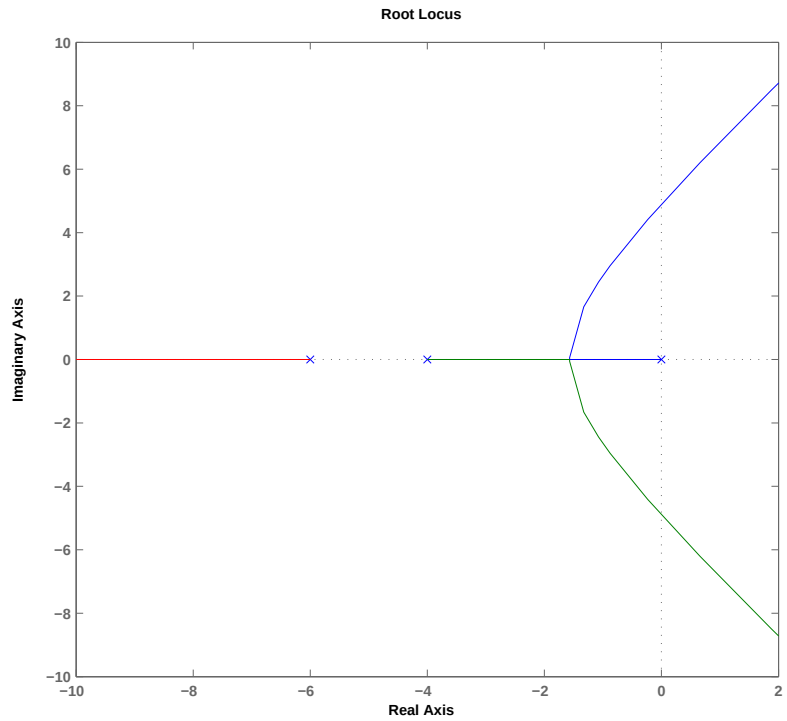
$$|s| = \sqrt{1 + 2^2} = \sqrt{5} = 2.236$$

Magnitude from Pole 2:

$$|s+4| = \sqrt{3^2 + 2^2} = \sqrt{13} = 3.6$$

Magnitude from Pole 3:

$$|s+6| = \sqrt{5^2 + 2^2} = \sqrt{29} = 5.385$$



$$k = \frac{|s-p_1| \cdots |s-p_n|}{|s-z_1| \cdots |s-z_m|} = \sqrt{5 \cdot 13 \cdot 29} = 43.4$$

Calculating Gain at a Point on the Root Locus
Numerical Example: A point NOT on the real axis

Points on the real axis are easiest. Lets try the point at $s \cong -1 + 2\imath$

$$G(s) = \frac{1}{s(s+4)(s+6)}$$

Magnitude from Pole 1:

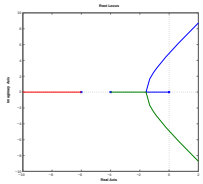
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$$k = \frac{|s-p_1|\cdots|s-p_n|}{|s-z_1|\cdots|s-z_m|} = \sqrt{5 \cdot 13 \cdot 29} = 43.4$$

The magnitude contributions at a complex point $s = \sigma_s + \imath\omega_s$ from pole $p = \sigma_p + \imath\omega_p$ are calculated from

$$|s-p| = \sqrt{(\sigma_s - \sigma_p)^2 + (\omega_s - \omega_p)^2}$$

Finding a Gain which meets Performance Specs

Numerical Example: Suspension System

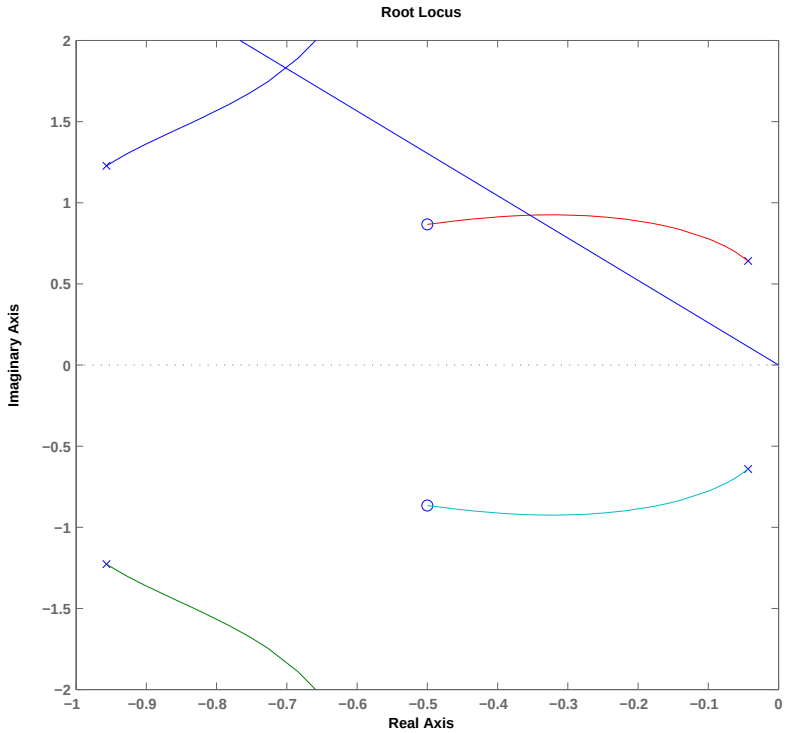
Can the Suspension system achieve 30% overshoot using proportional feedback?

$$G(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + s + 1}$$

- $M_P = .3$ defines the line at

$$\omega = \frac{\pi}{\ln(M_{p,desired})}\sigma$$

- Examine the gain at
  - $s_1 = -.3536 + .922i$
  - $s_2 = -.7 + 1.83i$

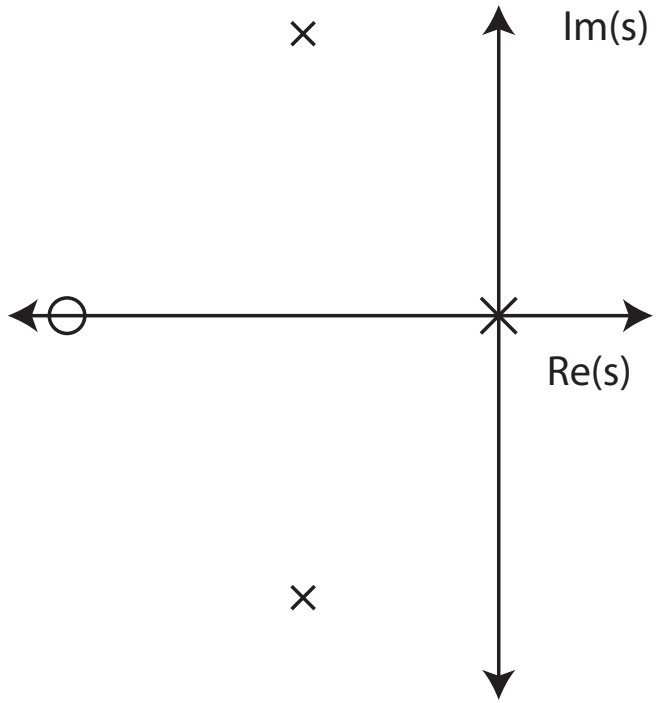


$$k(s_1) = 3.58$$
$$k(s_2) = 2.6153$$

Calculating Gain

DIY Example

$$G(s) = \frac{s+2}{s(s^2+2s+2)}$$



Find the gain for a point on the root locus (real axis) where $T_s = 8s$

Calculating Gain

Matlab

The Matlab syntax for root locus is

```
> rlocus(n,d)
```

where

- `n` is a vector of the coefficients of the numerator of G
- `d` is a vector of the coefficients of the denominator of G

Example:

$$G(s) = \frac{s^2 + 4s + 8}{s^6 + 2s^5 - s^3 + 2s^2 + 1}$$

- `> n = [1 4 8]`
- `> d = [1 2 0 -1 2 0 1]`

To find the gain at a point on the root locus:

- Plot the root locus.
- `> rlocfind(n,d)`
- Use the cursor to select the point.

Generalized Root Locus

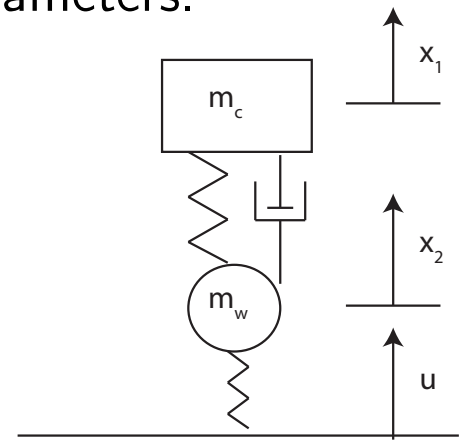
Visualizing the effect of changes in system parameters

We may want to know the effect of changing other parameters.

Examples:

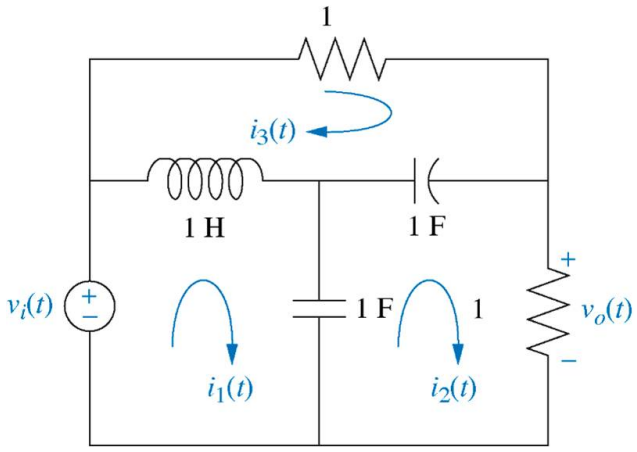
Physics (e.g. Suspension System)

- Spring Constant
- Damper
- Wheel Mass



Circuits (e.g. Toaster, Video Surveillance)

- Resisters
- Capacitors
- Inductors



Maybe there is no control at all!

Root Locus as a General Tool

What do parameters do?

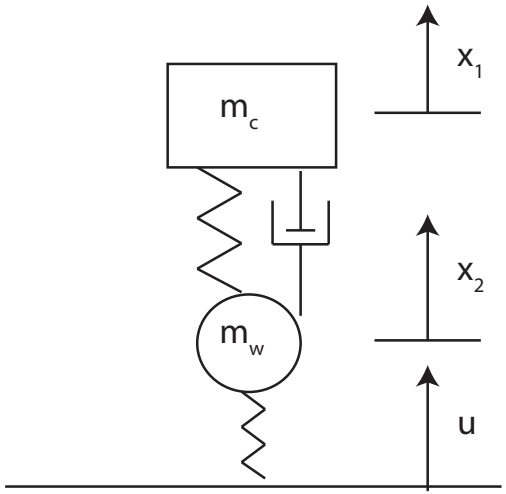
Suspension system: The full TF:

$$\frac{K_2(m_c s^2 + cs + K_1)}{m_c m_w s^4 + c(m_w + m_c)s^3 + (K_1 m_c + K_1 m_w + K_2 m_c)s^2 + cK_2 s + K_1 K_2}$$

The Effect of Damping Constant: c

- No Feedback
- Only examine c
  - Everything else is 1.

$$G(s) = \frac{s^2 + cs + 1}{s^4 + 2cs^3 + 3s^2 + cs + 1}$$



Where are the Poles?

$$s^4 + 2cs^3 + 3s^2 + cs + 1 = 0$$

$$s^4 + 3s^2 + 1 + c\,(2s^3 + s)$$

$$= d(s) + c\,n(s) = 0$$

Looks like the root locus!

Root Locus as a General Tool

What do parameters do?

$$G(s) = \frac{s^2 + cs + 1}{d(s) + c\,n(s)}$$

- $d(s) = s^4 + 3s^2 + 1$
- $n(s) = 2s^3 + s$

The root locus is the set of roots of

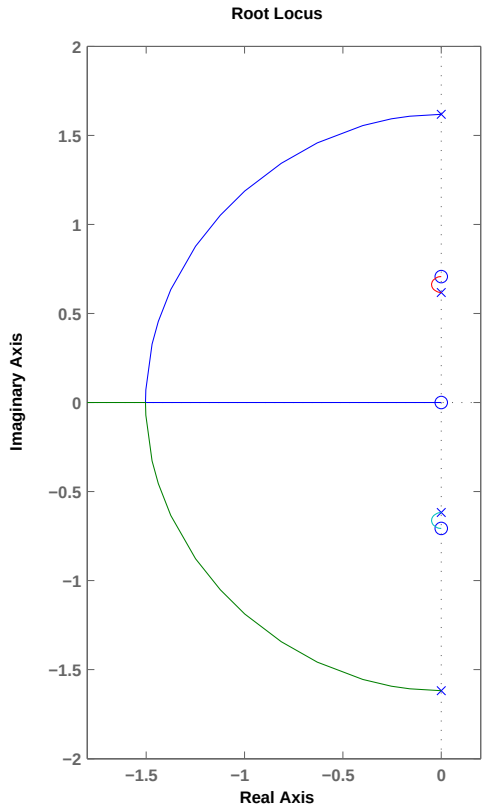
$$d(s) + kn(s)$$

We plot the root locus for

$$G_c(s) = \frac{n(s)}{d(s)} = \frac{2s^3 + s}{s^4 + 3s^2 + 1}$$

Note that G_c is totally fictional!

G_c must still be proper (n is lower degree than d).



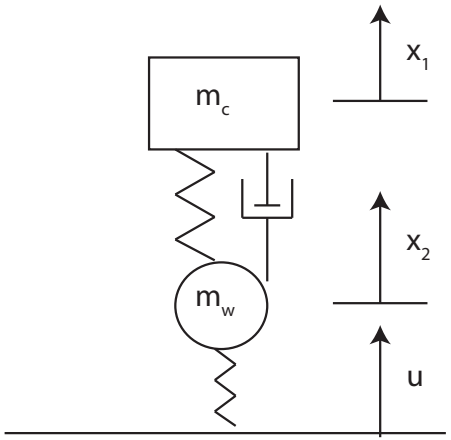
The effect of changing c is small.

Root Locus as a General Tool

Suspension Example: Damping Ratio

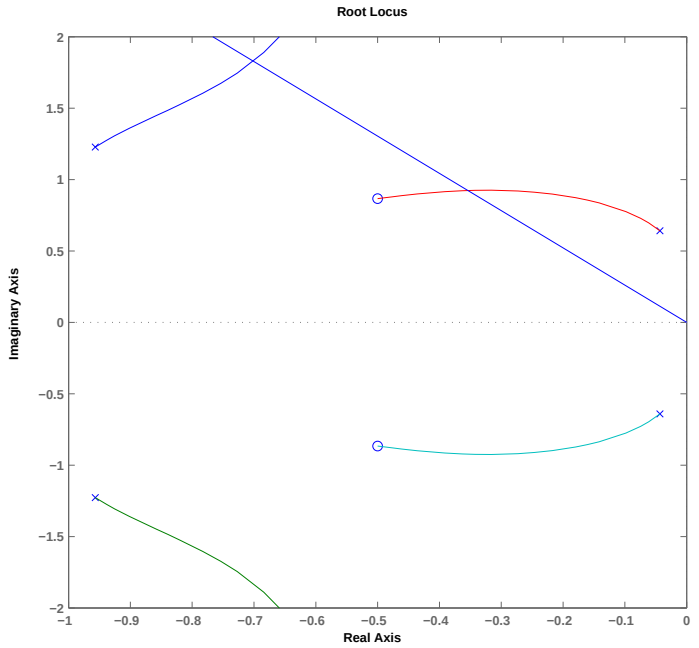
Root locus tells us:

- Changing c won't help with overshoot.
- We need Feedback!



Set $c = 1$ and plot the root locus

$$G(s) = \frac{s^2 + s + 1}{s^4 + 2s^3 + 3s^2 + s + 1}$$



- Examine the gain at
  - $s_1 = -.3536 + .922i$
  - $s_2 = -.7 + 1.83i$
- Find Crossover Points
  - $k = 3.58$
  - $k = 2.61$

We'll want $k \cong 3$.

Root Locus as a General Tool

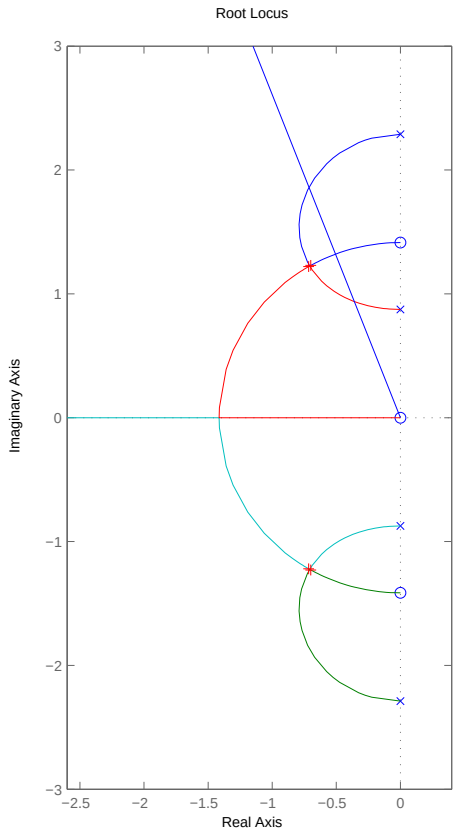
Suspension Example: Damping Ratio

Closed Loop Transfer Function:

$$\frac{kG(s)}{1+kG(s)} = \frac{k(s^2+cs+1)}{k(s^2+cs+1)+s^4+2cs^3+3s^2+cs+1}$$

Can damping ratio get us to 30% overshoot?

- With feedback
- Set $k = 3$ and plot root locus



Closed Loop Transfer Function ($k = 3$):

$$G_{kc}(s) = \frac{s^2+cs+1}{(s^4+6s^2+4)+c(2s^3+3s+s)}$$

Use `rlocfind` to pick off the best value of c.

Choose the point $s = -.71 + 1.215i$.

- Yields $c = \frac{|d(s)|}{|n(s)|} = 1.414$

Root Locus as a General Tool

Suspension Example: Damping Ratio

Using $c = 1.414$ and $k = 3$, the closed-loop transfer function is

$$\frac{kG_c(s)}{1 + kG_c(s)} = \frac{3s^2 + 4.24s + 3}{s^4 + 2.8s^3 + 6s^2 + 5.7s + 4}$$

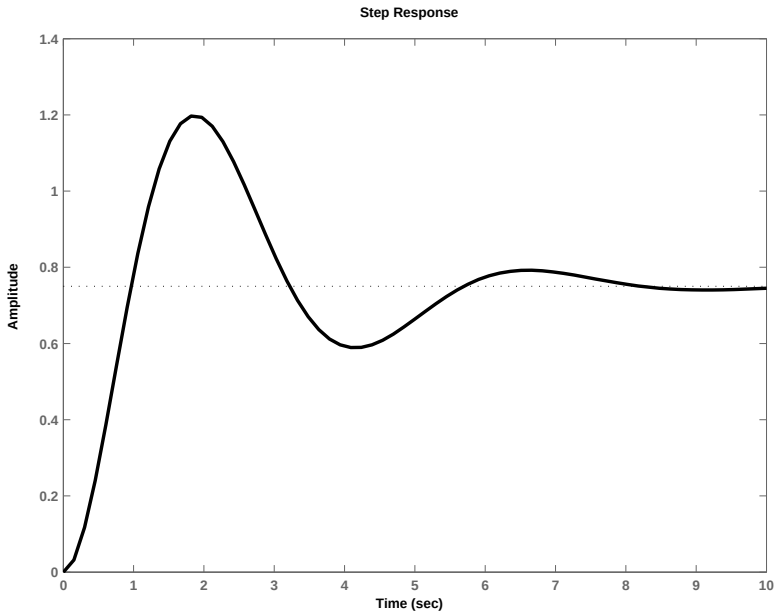
Which has repeated poles at

$$s_{1,2,3,4} = -.71 \pm 1.2i$$

Corresponds to an overshoot of

$$M_p = e^{\frac{\pi\sigma}{\omega}} = e^{-\frac{.71*\pi}{1.2}} = .18$$

The real overshoot is much bigger!

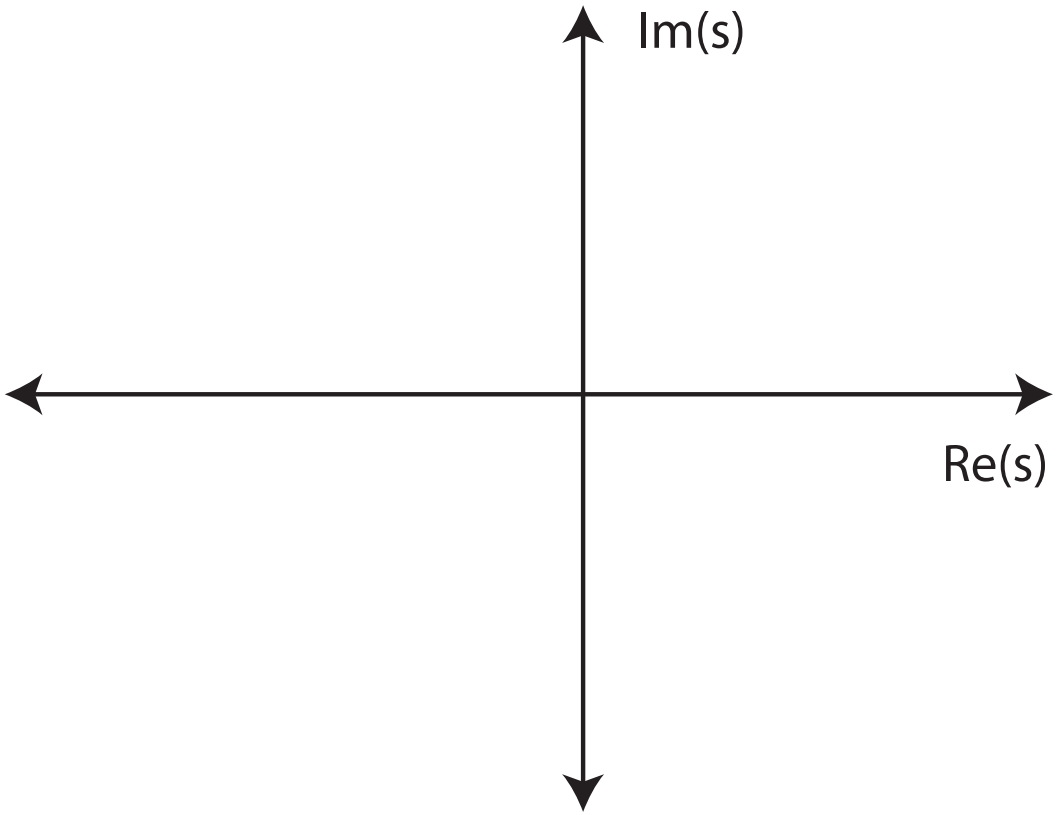


Root Locus as a General Tool

DIY Example

$$G(s) = \frac{s^2 + b^2 s + b}{s^3 + (7+b)s^2 + (12+b)s + b}$$

Find the optimal value of b.



Summary

What have we learned today?

Picking Points?

- Calculating the Gain
- Satisfying Performance Criteria

Generalized Root Locus?

- What about changing *OTHER* parameters
- T_D, T_I, et c.
- mass, damping, et c.

Next Lecture: Generalized Root Locus and Design Problems